

ANNEXURE II

SUBMISSION OF PROJECT SYNOPSIS AND GUIDE ACCEPTANCE FORM

(To be submitted to the Project Steering Committee)

PART A: Project Synopsis

I. Student Details:

1. Name of the Program: BCA
2. Name of the Student: SUTHAR DHRUVKUMAR NITINKUMAR
3. Roll Number: 2314506750
4. Session & Year: SEM 6 2026

II. Project Synopsis:

1. **Title of the Project:** Insectopedia: An Intelligent Digital Taxonomy and Information System for Entomological Classification using Python and Django
2. **Introduction and Review of Literature:** In the modern scientific community, accessing structured entomological data remains a challenge for students and hobbyists. While vast amounts of data exist in textbooks, they are often difficult to navigate or search efficiently in a digital format. This project aims to develop a robust, web-based "Insectopedia" that automates the organization of species data into an intuitive digital repository
 - **Problem Statement:** Traditional entomological records are often stored in static formats that lack cross-referencing and rapid searchability, leading to "Information Fatigue" for researchers. Small educational institutions often lack the budget for high-end biological database systems. There is a critical

need for a lightweight, accurate, and easy-to-deploy tool that bridges the gap between raw scientific data and user-friendly digital accessibility.

- **Literature Review:** Existing systems often rely on simple alphabetical lists, which fail to represent the complex hierarchical nature of insect taxonomy (Family, Genus, Species). Recent advancements in web frameworks allow for relational database management that can mirror these biological hierarchies. Studies show that interactive digital classification systems can increase student engagement and data retention by over 60%. By utilizing Python-based logic, we can move beyond static pages to dynamic "Entity Relationships," identifying specific traits and habitats across different insect families accurately.

3. **Objectives of the Study:** The primary objectives are:

- a. To develop a Django-powered system that categorizes and stores extensive data on various insect families.
- b. To implement a robust search and filter algorithm that allows users to find species based on specific taxonomic traits.
- c. To provide a responsive UI/UX dashboard for enthusiasts to explore all categories and contribute to the digital archive.
- d. To implement an educational feedback loop by suggesting related species based on user search history.

4. **Research Methodology:** The project will follow the Agile Development Model, allowing for iterative improvements across four phases:

- a. **Phase 1: Data Structuring:** Using Django Models to define the relationship between insect Families, Genera, and Species.

- b. **Phase 2:** Backend Logic: Implementing the "python manage.py" architecture to handle server-side requests and database migrations.
 - c. **Phase 3:** Search Optimization: Developing a Python-based search logic to calculate matches between user queries and stored species data.
 - d. **Phase 4:** Frontend Integration: Designing a reactive web interface where the search bar and categorical "Explore All" features are integrated into a seamless workflow..
5. Technology to be Used: Python 3.x – Chosen for its robust backend capabilities and ease of data management
- a. **Web Framework:** Django – An industry-standard framework for building secure and scalable web applications
 - b. **Database: PostgreSQL/SQLite** – For organizing complex taxonomic data into relational tables.
 - c. **Frontend UI/UX:** HTML5, CSS3, and JavaScript – For creating a responsive and interactive user experience.
 - d. **Development Environment:** VS Code and Git – For managing code versions and debugging.
6. **References: Books & Academic Papers:**
- a. Django Software Foundation. (2026). Django Documentation: Models and Databases. Available at: <https://docs.djangoproject.com>.
 - b. Sarker, I. H. (2021). Machine Learning & Data Structures: Real-World Applications. SN Computer Science.
 - c. Python Software Foundation. Python Language Reference, version 3.x. Available at: <https://www.python.org>.
 - d. Kaggle. Entomology and Biodiversity Datasets. Available at: <https://www.kaggle.com/datasets>.

III. Guide Details:

1. Name of Proposed Guide: Mr. PRAMOD MATHUR
2. Guide Registration No. (If available):
3. Designation:
4. Affiliation:
5. Qualification:
6. Total Experience:
7. Communication Address:
8. Contact No.:
9. E-mail ID:

PART – B: Guide Acceptance

I, Dr./Mr./Mrs. _____ working with
_____ as _____ hereby
confirm my willingness to guide Mr./ Ms.
_____ Reg No. _____ for the topic
(title of the project) _____
_____ during the period _____ (month/year) to _____
(month/year).

Place:

Date:

(Signature of the Guide)

(Note: A Guide needs to get registered with the University if he/ she is guiding a MUJDOE project for the first time. The guide Registration form can be downloaded from the LMS portal)

DECLARATION

I hereby declare that this project synopsis is an original work carried out by me and has not been/will not be submitted to any other University for the fulfilment of any course of study.

Place: Airforce Camp, Panagarh.

Date: 25 Apr 26

(*Filled in application forms to be signed by both student and the Guide. Forms must be scanned in either .pdf/.doc format and submitted through the LMS student's Login.)

A handwritten signature in blue ink, appearing to read 'Dhruv', is written over a horizontal line.

(Signature of the Student)